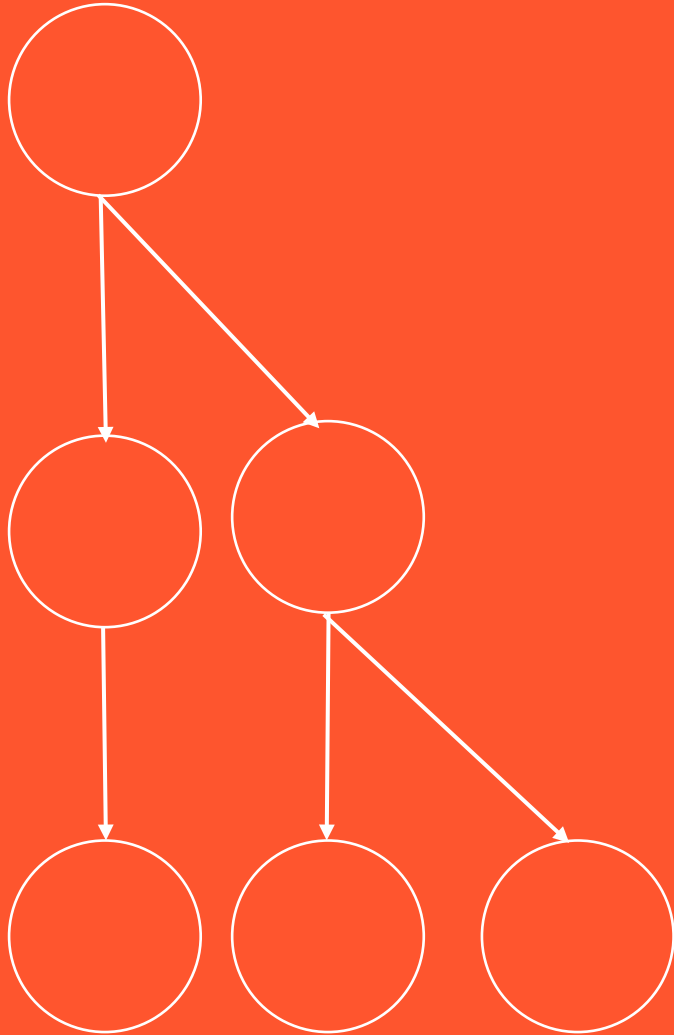


BCOG 100

Language

Lecture 2: The Psychology of Language



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 11-1

Password: chomsky

Question: The article discusses language as being "hierarchical". What does that mean?

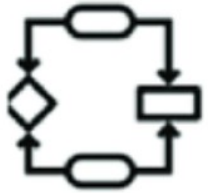


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Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

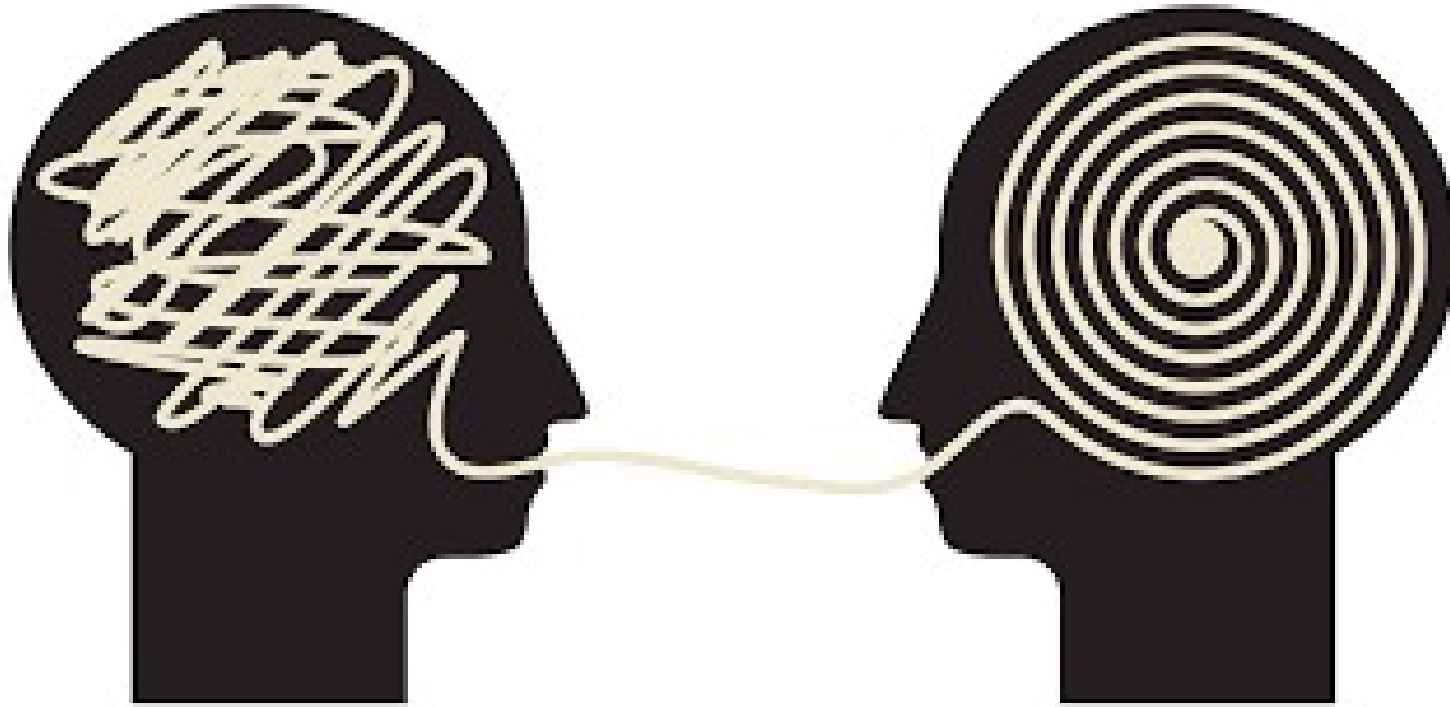
Last Time:

Why language may have evolved, and what made it unique and beneficial

Today:

How minds represent, learn, and use language.

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How Can a Mind Do Language?

- Language seems infinite, but the brain is finite
- We produce and understand sentences we've never heard
- How can a physical mind generate such creativity?

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Skinner's Verbal Behavior

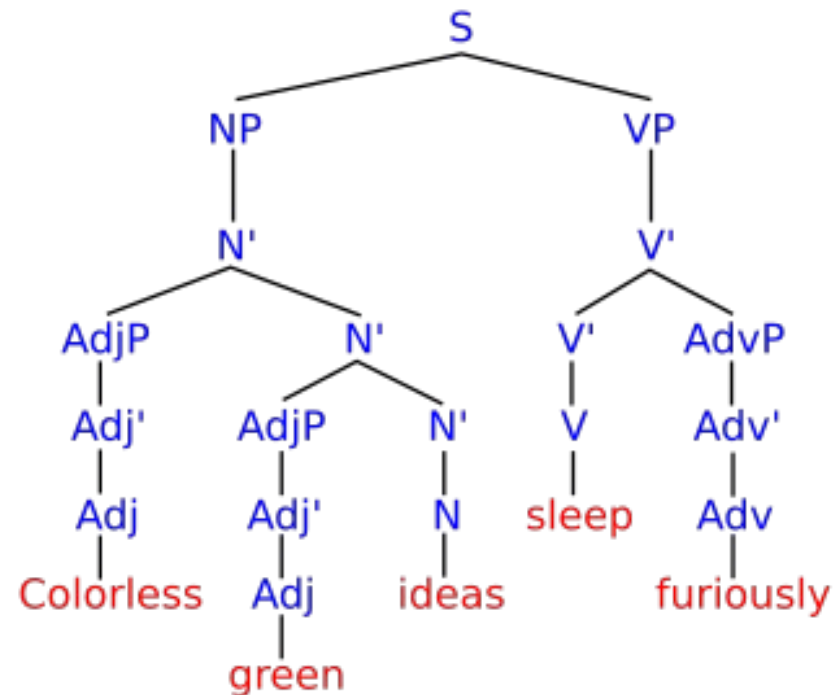
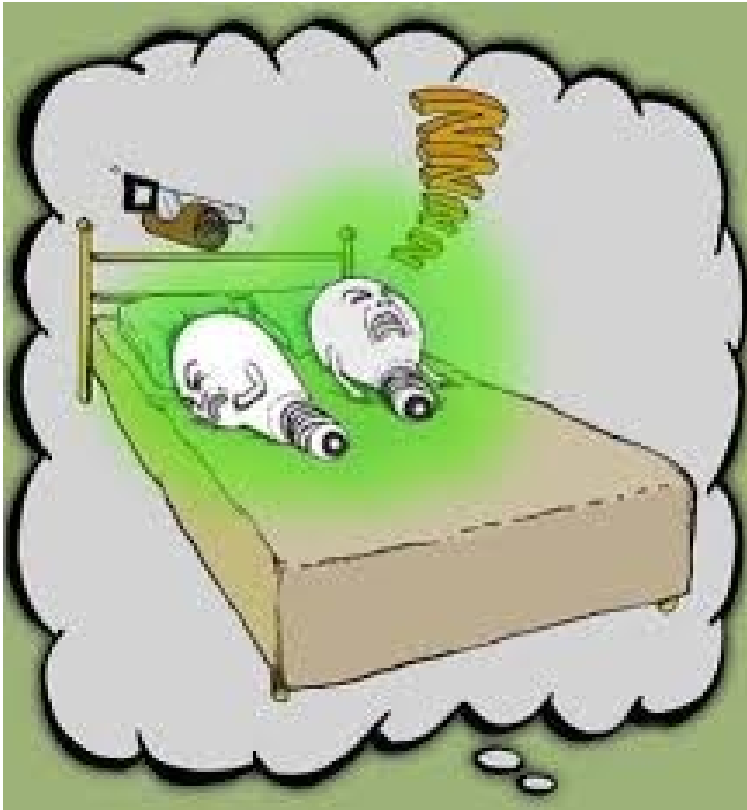
- Language as learned behavior through reinforcement
- Speech shaped by environment, not internal rules
- A powerful idea that met powerful criticism

Verbal Operants



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Ideas colorless sleep green furiously
Colorless green ideas sleep furiously



Chomsky's Critique of Behaviorism

- According to Chomsky, language is creative, not just imitative
- Children say things they've never heard
- Reinforcement can't explain infinite productivity

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Universal Grammar and the Birth of Cognitive Science

- Chomsky argues humans are born with an innate language faculty
- Universal Grammar constrains what languages can be
- Language became the model system for studying the mind

Sub-Idea	Brief Description
Universal Grammar (UG)	A set of innate linguistic principles shared by all humans that constrain what forms a possible language can take.
Parameters and Triggers	Children “set” parameters (e.g., word order, subject omission) based on limited exposure to their native language.
Poverty of the Stimulus	The input children receive is too limited and error-filled to explain their rapid mastery, implying innate knowledge.
Deep vs. Surface Structure	Sentences have an abstract underlying structure (deep) that can be transformed into various surface word orders.
Transformational Rules	Grammar includes operations (e.g., move the auxiliary for a question) that link related sentence forms.
Competence vs. Performance	UG describes the idealized mental grammar (competence) distinct from everyday errors in speech (performance).
Language Faculty / LAD	A specialized, evolutionarily evolved system in the brain (Language Acquisition Device) for building grammar.

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Grammar

$S \rightarrow NP VP$

$NP \rightarrow (AP) Det N$

$VP \rightarrow V (NP) (PP)$

$PP \rightarrow P NP$

$AP \rightarrow A$

$Det \rightarrow [the, a, my, your]$

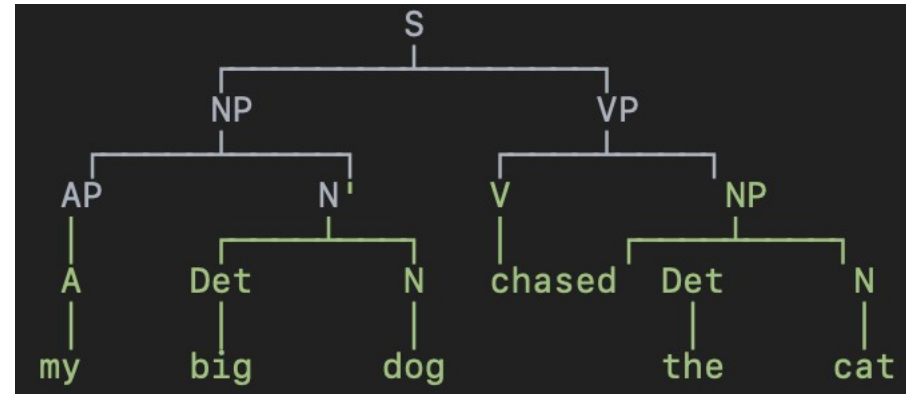
$N \rightarrow [dog, cat, friend, book]$

$V \rightarrow [chased, read, saw, likes]$

$P \rightarrow [with, to, near]$

$A \rightarrow [big, small, happy, tired]$

Syntactic Structure



The Mind as a Rule System

- Language as computation over symbols
- Sentences generated by rules, not memorized strings
- The mind as a grammar engine for thought

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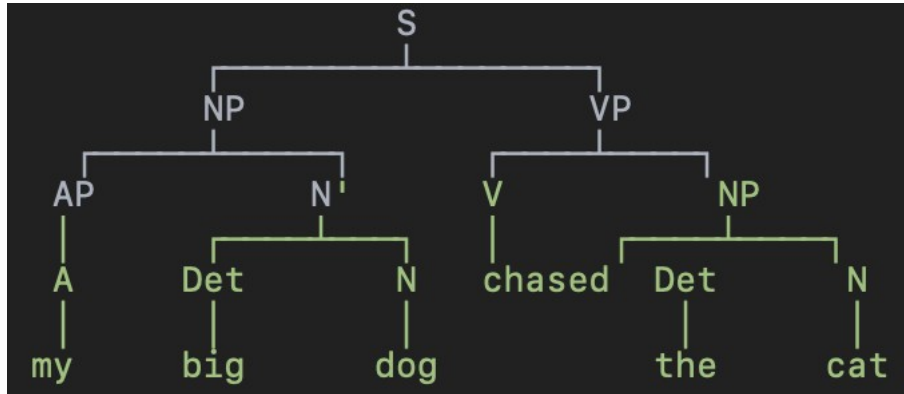
Grammar in the Mind

- Sentences unfold in structure, not in a straight line
- The mind tracks categories, dependencies, and expectations
- Psycholinguistics reveals grammar as a mental process

Garden-Path Sentences

- The old man the boats.
- The horse raced past the barn fell.
- While Anna dressed the baby played in the crib.
- The man whistling tunes pianos.
- The spy saw the man with binoculars.

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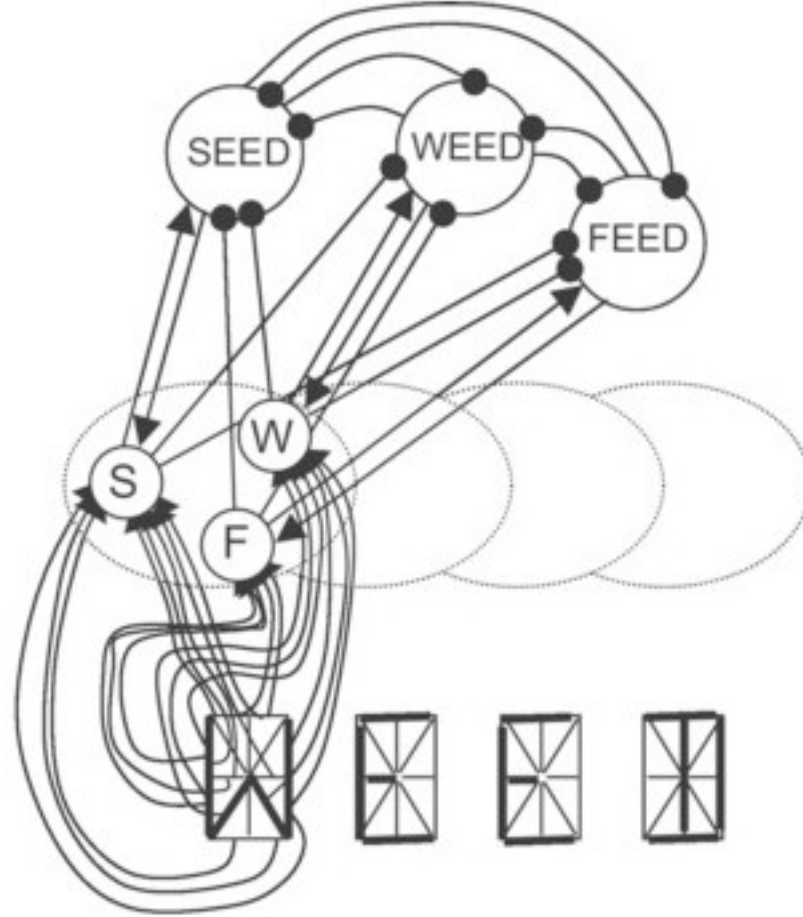
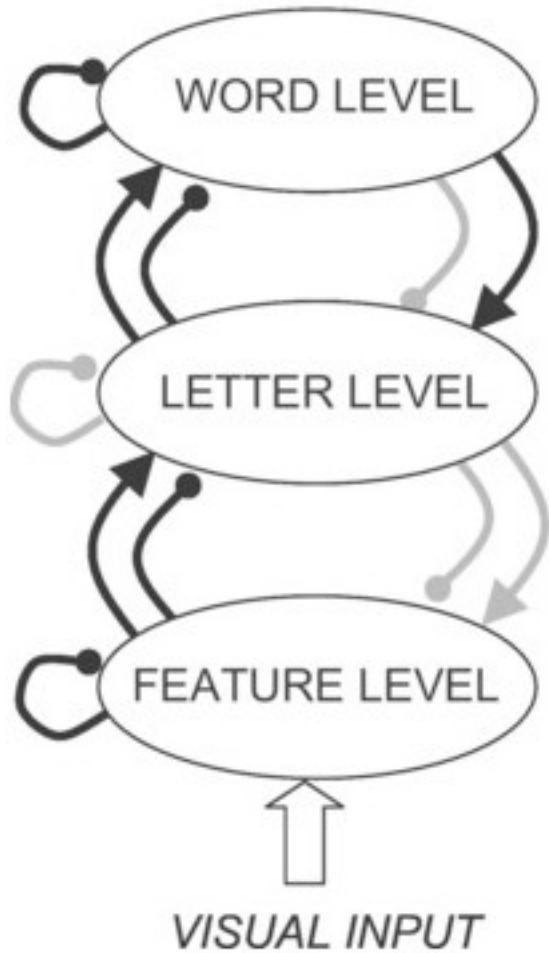


the ... uh.. big... no I mean small dog chased the cat

Limits and Critiques of the Symbolic Model

- Rules explain structure, but not learning
- Real language is messy, graded, and contextual
- The symbolic mind may be too rigid to capture it

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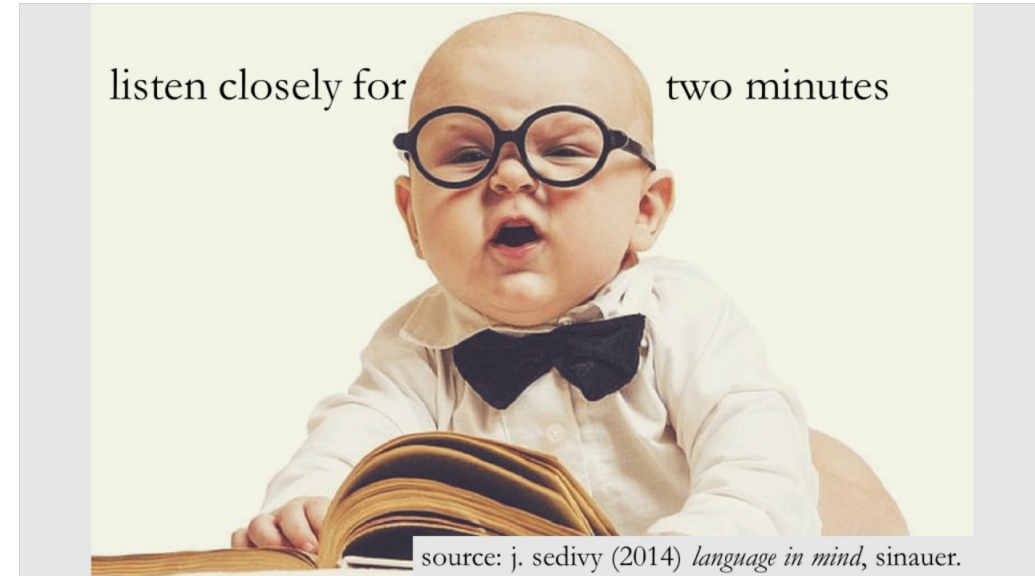
Connectionism and the Return of Learning

- Structure can emerge from experience
- Networks learn patterns, not explicit rules
- Generativity may come from prediction, not innateness

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Statistical Learning and the Structure of the Input

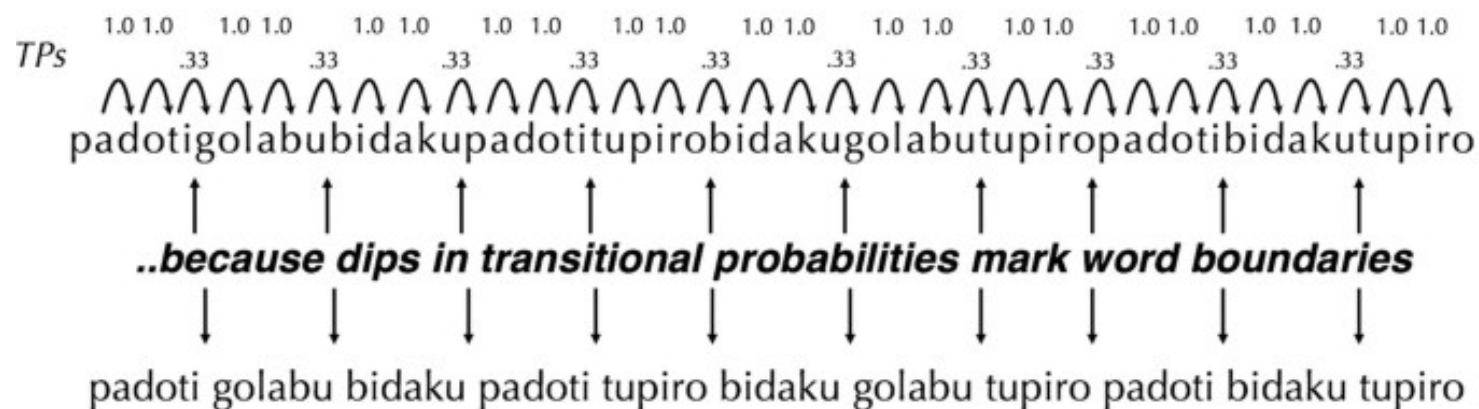
- Learners track patterns in what they hear
- Structure can be discovered in the input itself
- The world may teach more than Chomsky thought



A continuous stream of syllables...

“padotigolabubidakupadotitupirobidakugolabutupiropadotibidakutupiro”

..can be segmented into units from statistical cues...



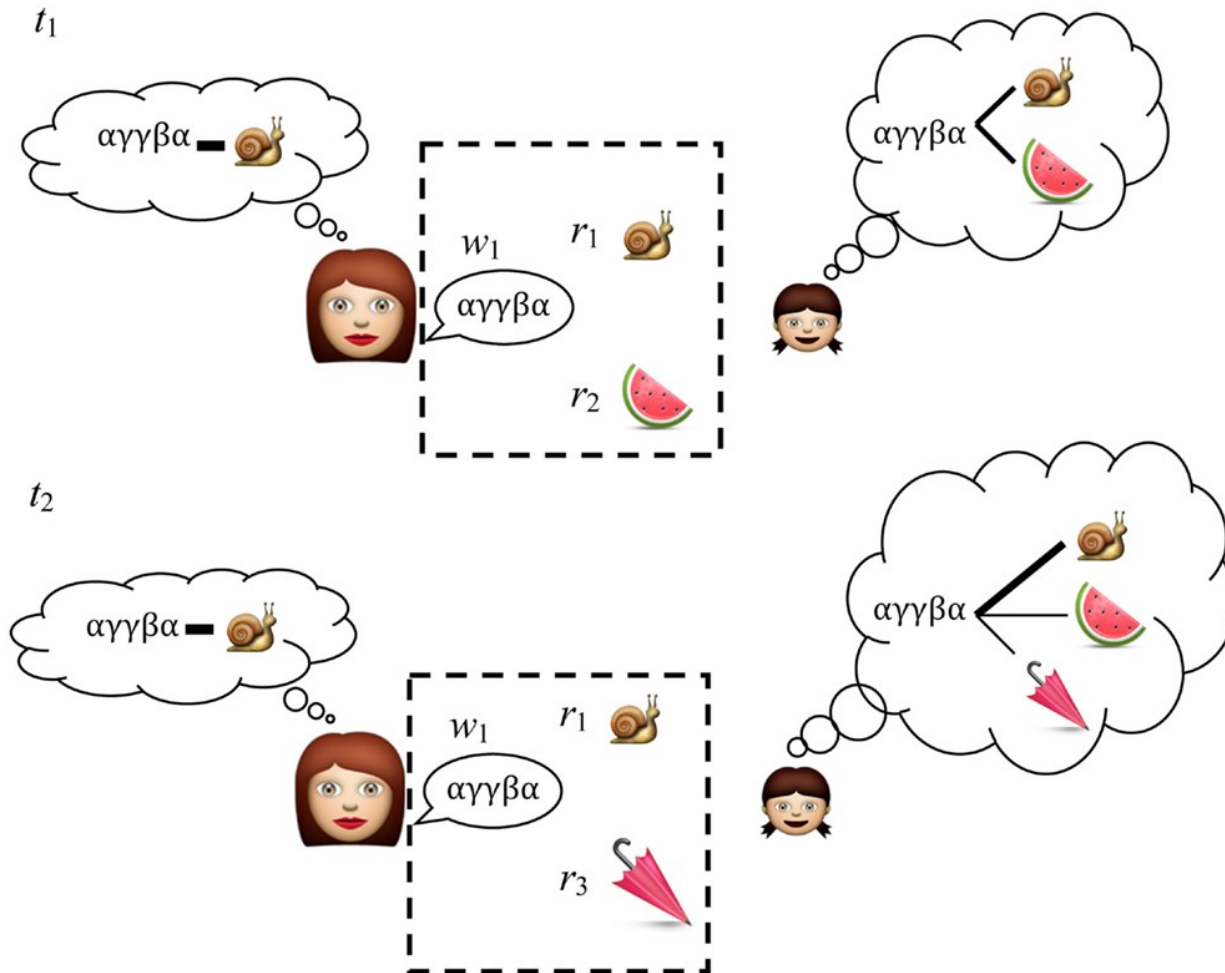
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The Social Prerequisites for Language

- Humans are driven to share minds, not just sounds
- Language depends on motivation to communicate
- Sociality may be the key difference, not intelligence

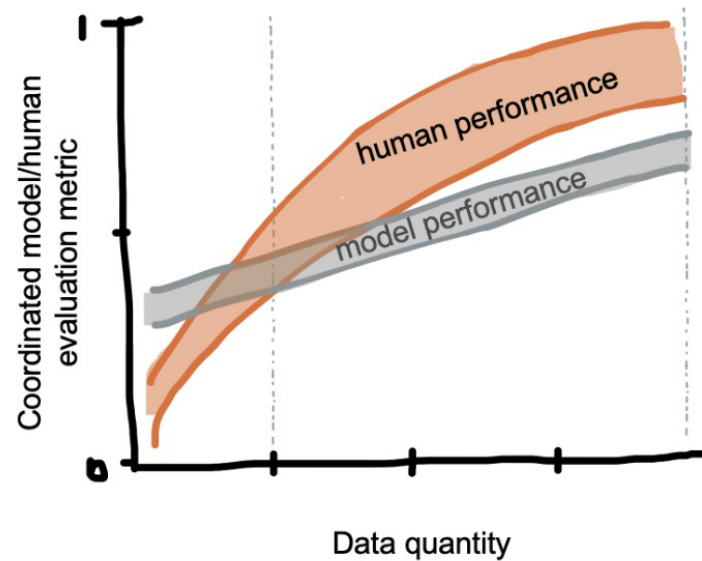
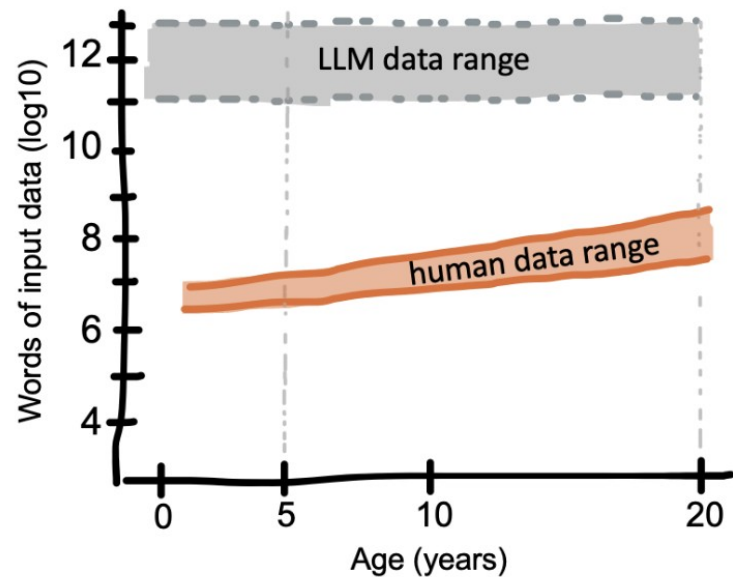
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Usage-Based and Constructivist Approaches

- Language grows out of social use
- Learning, attention, and intention work together
- Grammar emerges where communication succeeds

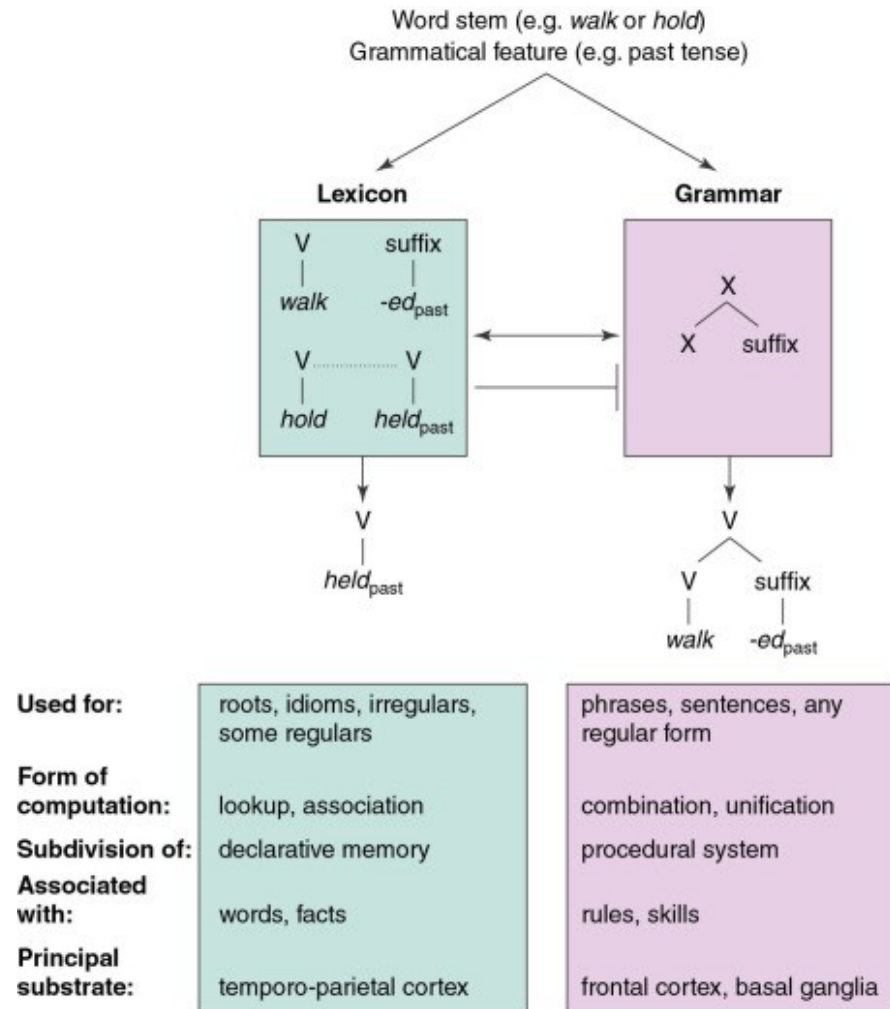
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Modern Machine Learning and Language

- Computers now learn language from data, not rules
- Patterns emerge from prediction, not programming
- Machines echo human debates about learning and structure

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TRENDS in Cognitive Sciences

Debate Example 1: Morphology Acquisition

- Do we apply rules or learn patterns?
- Past tense and plurals reveal how we generalize
- Language learning turns memory into structure

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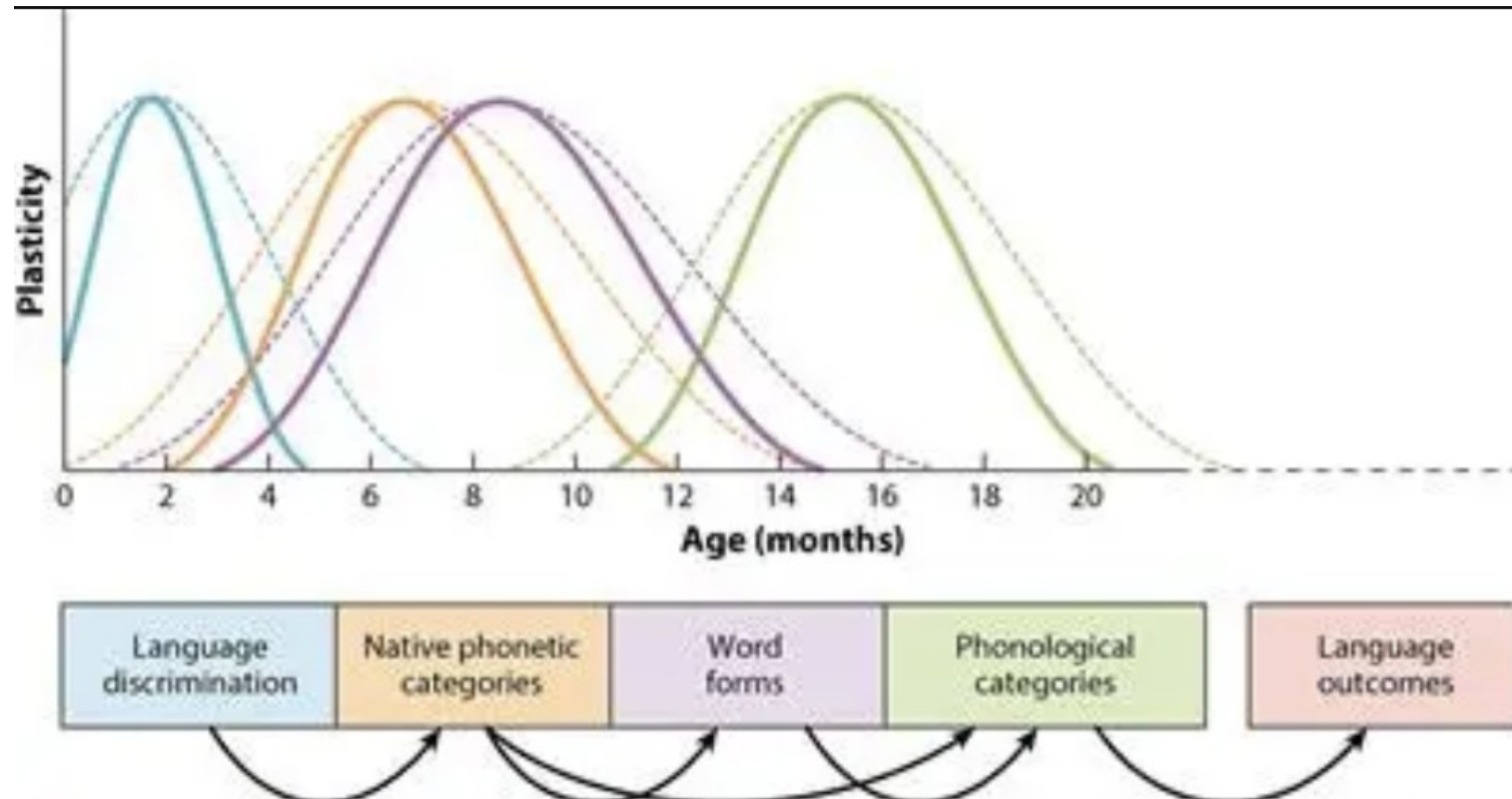
B

The man will	ride	the	motorbike.
The girl will	ride	the	carousel.
The man will	taste	the	beer.
The girl will	taste	the	sweets.

Debate Example 2: Understanding Sentences: Rules or Predictions?

- Do we follow grammar or guess ahead?
- Meaning unfolds through expectation and context
- Parsing may be prediction in disguise

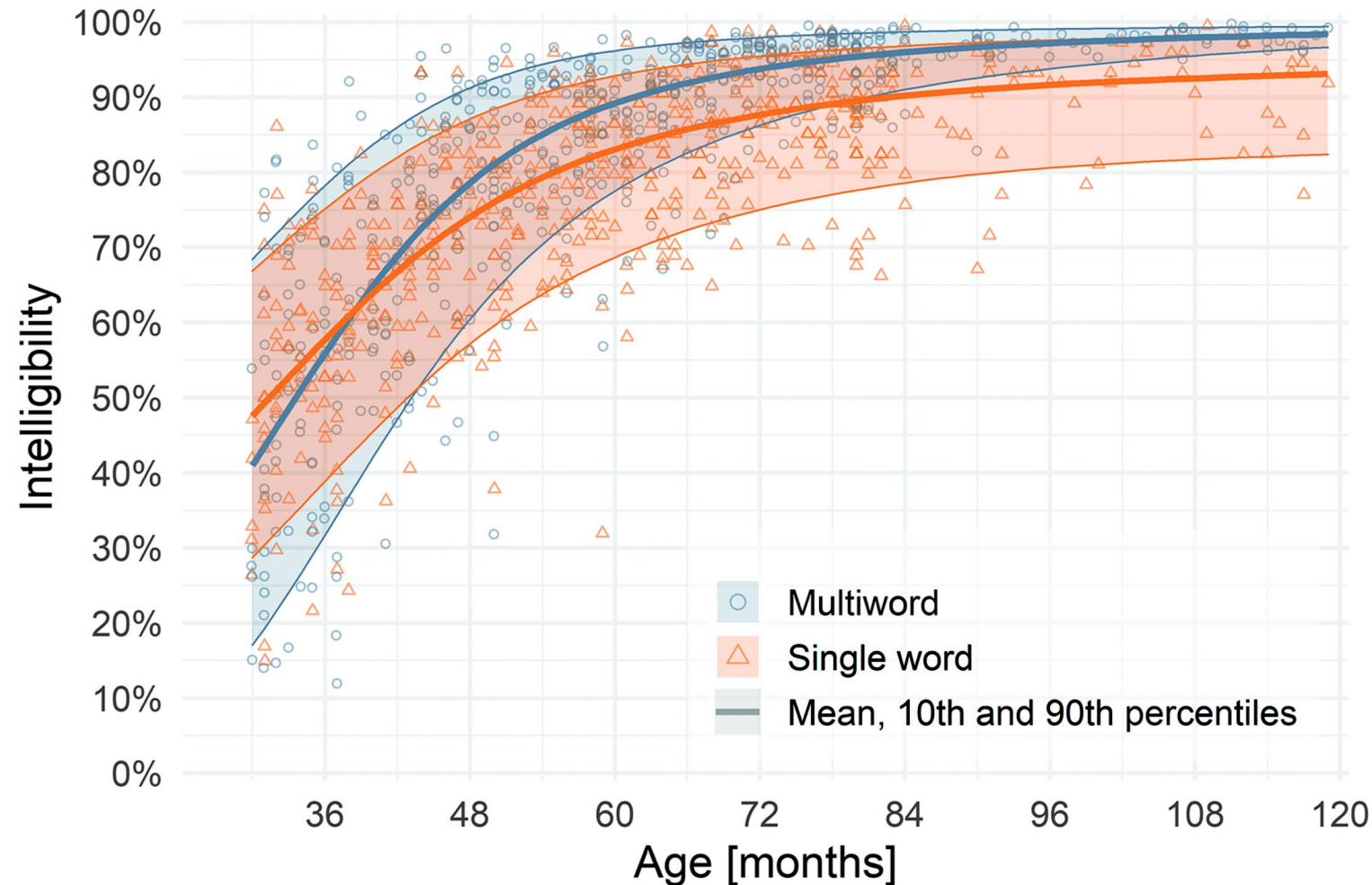
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Debate Example 3: Critical Periods in Language Acquisition?

- Is there a window for mastering language?
- Does biology close the door, or can experience reopen it?
- Timing tests the balance of nature and nurture

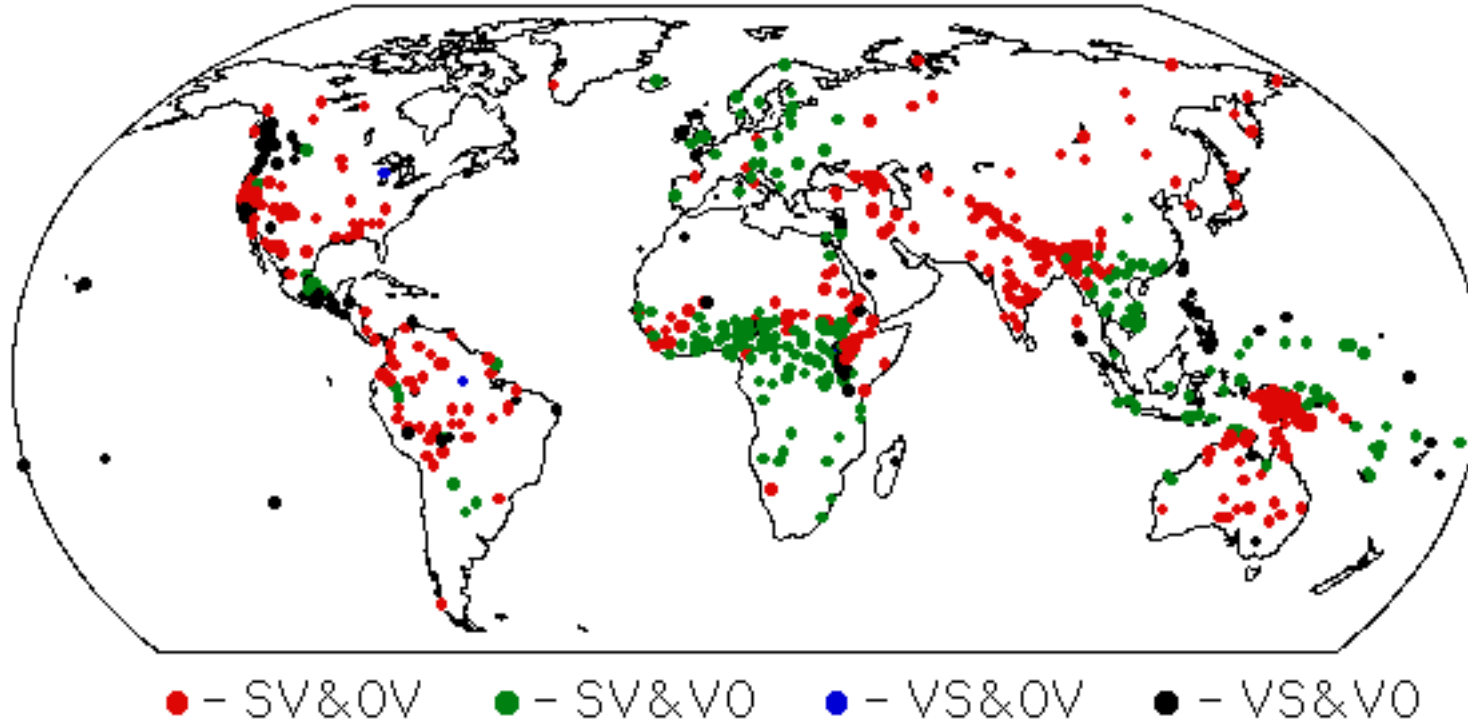
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Debate Example 4: How Do Children Learn Words So Fast?

- New words learned in a single glance
- Innate biases or learned associations?
- Speed reveals the mind's hidden structure

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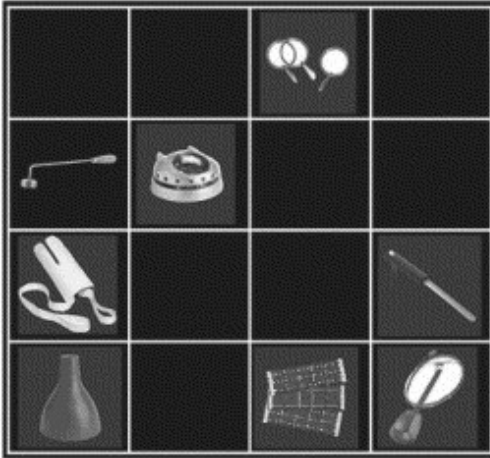


Debate Example 5: Word-Order Universals and Typology

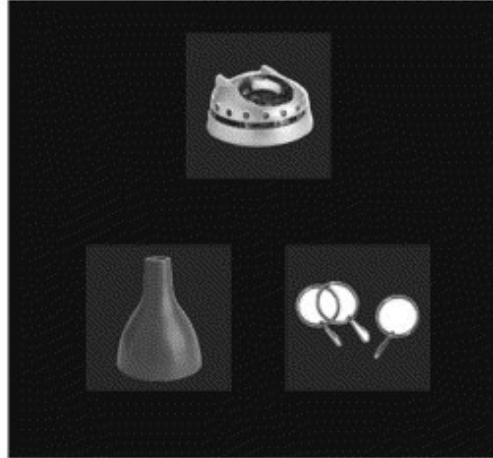
- Are all languages built on shared rules?
- Universal Grammar or cultural adaptation?
- Patterns may reflect processing, not prescription

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Experiment 1



Experiment 2

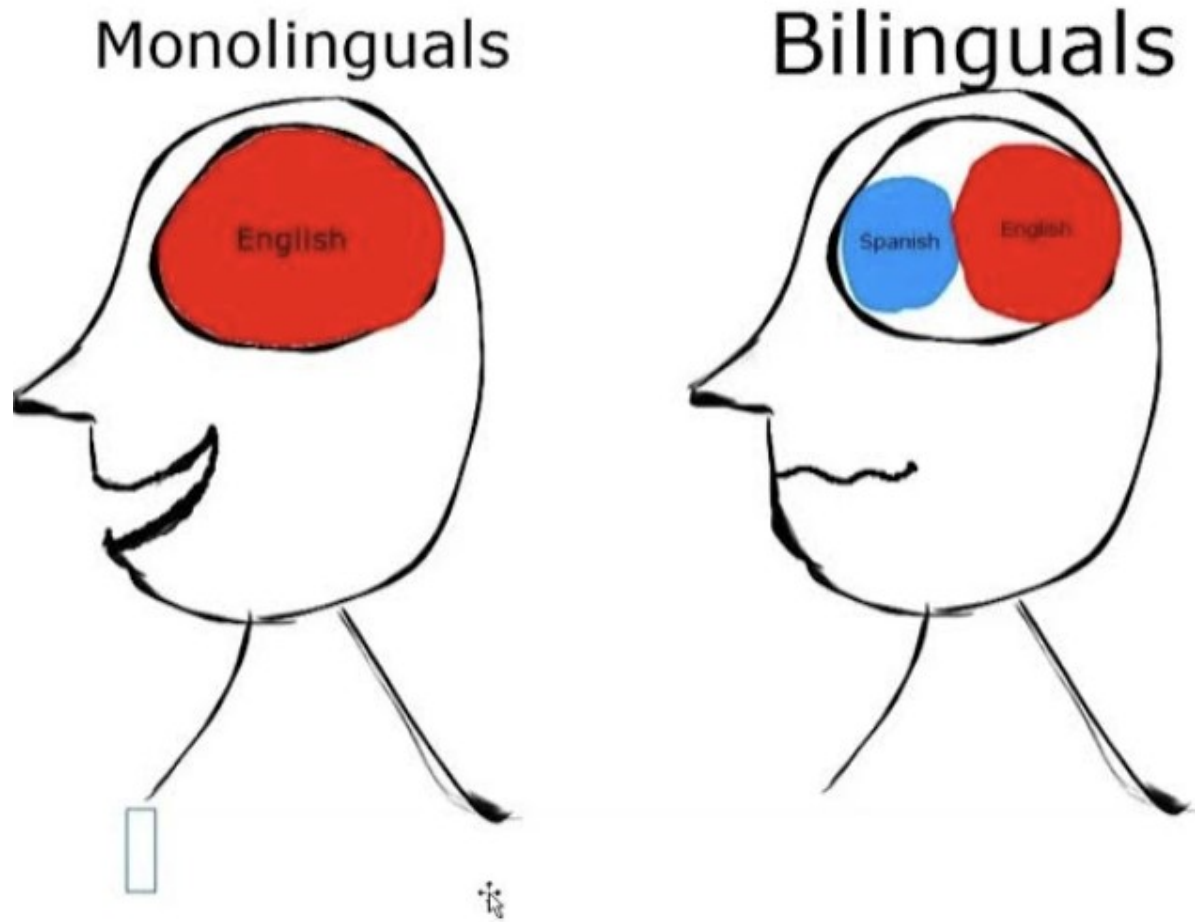


Condition	First Mention	Second Mention	Target Instruction
Same Speaker-Maintain	Black vase	Black vase	Black vase
Same Speaker-Break	Upside-down funnel	Upside-down funnel	Black vase
Diff. Speaker-Maintain	<i>Black vase</i>	<i>Black vase</i>	Black vase
Diff. Speaker-Break	<i>Upside-down funnel</i>	<i>Upside-down funnel</i>	Black vase

Debate Example 6: Pragmatics and Theory of Mind

- Language means more than it says
- Do we need a special module for intention?
- Understanding minds may underlie understanding words

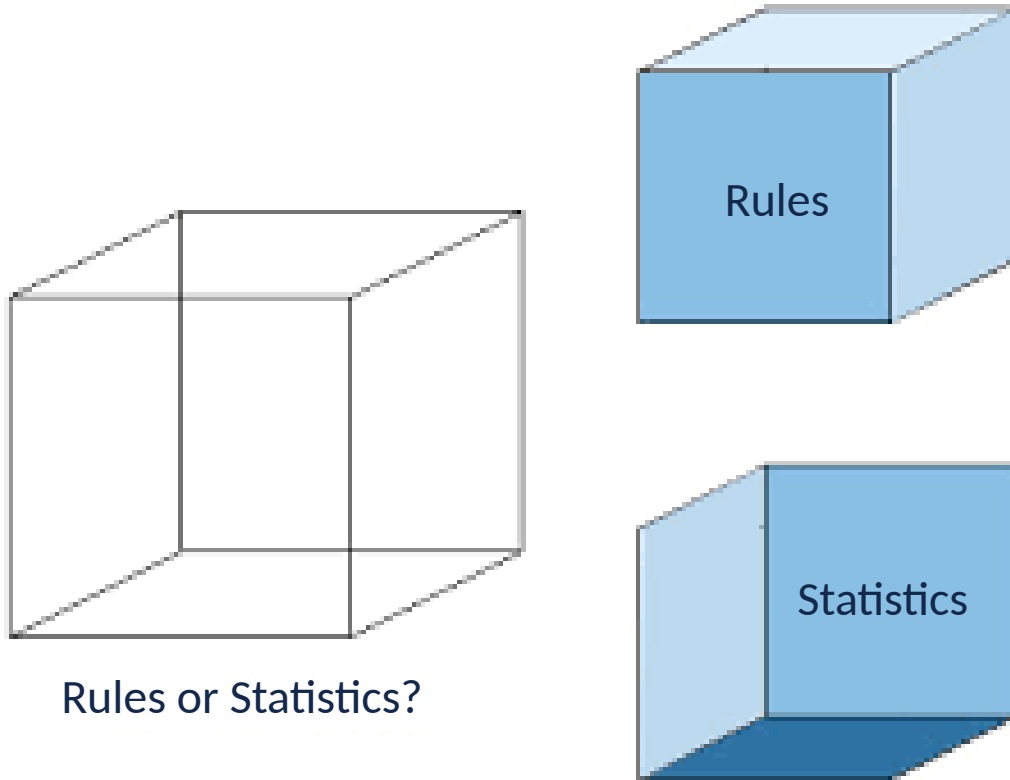
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Debate Example 7: Bilingualism and Cross- Linguistic Influence

- Two languages, one mind
- Does each follow fixed rules or share common ground?
- Flexibility may reveal how language adapts

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Integrating Theories: Hybrid Models

- Rules and learning each explain part of the story
- Structure and experience work hand in hand
- Language may be both built and built-upon

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Other Major Questions in the Psychology of Language

- Language connects to every corner of cognition
- How it works reveals how the mind works
- The questions keep expanding, not ending

Debate

Language and Thought

Modularity vs. Integration

Embodiment of Meaning

Innateness vs. Cultural Evolution

Inner Speech and Cognitive Control

Core Question

Does the language we speak shape how we perceive and reason, or does thought exist independently of language?

Is language a specialized cognitive module or a distributed system that shares resources with memory, attention, and perception?

Are words and concepts grounded in sensorimotor experience, or represented abstractly and amodally in the mind?

Which aspects of linguistic structure reflect biological design, and which emerge through cultural transmission and learning?

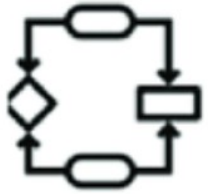
Does language function as a tool for self-regulation and planning, or is inner speech just a by-product of thought?

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Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Summary

- From structure to learning to meaning
- Language as prediction, communication, and thought

Next Time

- How language actually works in the mind